

1. REF5025 has fixed 2.5V output
2. Max output current is 10mA

12.1 Layout Guidelines

Figure 31 illustrates an example of a PCB layout for a data acquisition system using the REF4132-Q1. Some key considerations are:

- Connect low-ESR, 0.1-μF ceramic bypass capacitors at V_{DD} , V_{REF} of the REF4132-Q1.
- Decouple other active devices in the system per the device specifications.
- Using a solid ground plane helps distribute heat and reduces electromagnetic interference (EMI) noise pickup.
- Place the external components as close to the device as possible. This configuration prevents parasitic errors (such as the Sestbeck effect) from occurring.
- Do not run sensitive analog traces in parallel with digital traces. Avoid crossing digital and analog traces if possible, and only make perpendicular crossings when absolutely necessary.

12.2 Layout Example

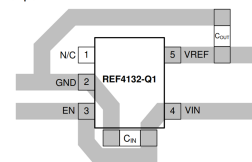


Figure 31. Layout Example

10.2.1 Design Requirements

A detailed design procedure is described based on a design example. For this design example, use the parameters listed in Table 2 as the input parameters.

Table 2. Design Example Parameters

DESIGN PARAMETER	VALUE
Input voltage V_{DD}	5 V
Output voltage V_{OUT}	2.5 V
REF4132-Q1 input capacitor	1 μF
REF4132-Q1 output capacitor	10 μF

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